

# Sample Paper

*This sample paper aims to indicate the type of questions and format that you will see in your final exam. The mark distribution and difficulty may be different.*

# **The Chinese University of Hong Kong**

## **Final Examination**

### **UGED1111 Logic 邏輯**

**Student Name:** \_\_\_\_\_  
**Student ID:** \_\_\_\_\_  
**Time Allowed:** 90 minutes  
**Total Marks:** 100

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#### **INSTRUCTIONS TO CANDIDATES**

- Put all your answers on the answer book.
  - Answer written on the question papers will not be marked.
  - There are six sections in total.
  - Assessment weighting: 40% of the whole course.
  - You may use the blank area of the question papers to write drafts.
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### Section I. True/False Questions (24 marks)

Put “T” for true statements and “F” for false statements. (2 marks for each; 2 mark will be deducted for incorrect answer) [W, Y are simple statements.]

1. It is possible for an inductive argument to have all true premises and a false conclusion.
2. If something is logical impossible, it is not technologically possible.
3. From the Aristotelian standpoint, “Some dogs are cute” is true if “Every dogs are cute” is true,
4. For any statement  $S_1$ ,  $S_2$ , and  $S_3$ , if  $S_1$  and  $S_2$  are contradictory, and  $S_2$  and  $S_3$  are logically equivalent, then  $S_1$  and  $S_3$  must be contradictory.
5. An argument with 2 true premises and 2 false premise is 50% sound.
6. By method of difference, we find the cause of an effect by looking for co-variation of two factors.
7. A cogent argument must be a strong a argument.
8. An argument from analogy has a true conclusion if it has all true premises.
9. A compound statement may not have a minor operator.
10. We committed the fallacy of begging the question when we ask the arguer a question.
11. If “ $W \supset Y$ ” is true, then it must be W causes Y.
12. We cannot have a contingent statement as a premise of an argument.

### Section II. Categorical Logic (8 marks)

Translate the following arguments into standard form categorical syllogism and use Venn diagrams to determine whether it is valid or invalid. (8 marks for each)

1. Not all politicians are corrupted. Only greedy persons are corrupted. Therefore, it is not the case that all politicians are not greedy.

### Section III. Indirect Truth Table (8 marks)

Use the indirect truth table to determine whether it is valid or invalid.

1.  $(A \vee B) \equiv C / \sim A \cdot C // \sim(B \supset \sim D)$

#### Section IV. Complete Truth Table (8 marks)

Use a complete truth table to determine whether the following propositions are logically equivalent, contradictory, consistent, and/or inconsistent. (0.5 mark will be deducted for any wrong or blank entry in your answer)

1.  $(P \vee Q) \cdot (Q \supset R)$  and  $(P \equiv Q) \supset R$

#### Section V. Informal Fallacy (20 marks)

Identify the fallacies committed by the following passage and give a brief explanation for your answer.

(1) 收集了 2000 名「兔兔」支持者的意見，全數認為「兔兔」是最可愛的，由此可見全世界的人都認為「兔兔」是最可愛的。

(2) 億萬富翁都會擁有自己的汽車。所以讓一個人成為億萬富翁的最佳方法是給他一輛車。

(3) 父母對讀中學的孩子說：你作為中學生不應該戀愛。如果你不同意，你以後一分錢零用錢也沒有。

(4) Tom's theory on education can't be right. He is a smoker and he has tattoo on his both arms.

#### Section VI. Natural Deduction (32 marks)

Use natural deduction to construct a formal proof of validity of the following arguments.

1.  $(A \vee B) \supset B$  //  $A \supset (A \supset B)$  (6 marks)

2.  $(C \vee D) \supset H$   
 $(D \supset H) \supset F$  // F (10 marks)

3.  $(N \supset M) \supset M$   
 $\sim M$  // N (4 marks)

4.  $\sim A \supset (\sim B \cdot M)$   
 $B \vee (M \cdot N)$   
 $A \supset N$  // NVM (12 marks)

## 8 RULES OF IMPLICATION

### 1. Modus Ponens (MP)

$$\begin{array}{l} p \supset q \\ p \\ \therefore q \end{array}$$

### 2. Modus Tollens (MT)

$$\begin{array}{l} p \supset q \\ \sim q \\ \therefore \sim p \end{array}$$

### 3. Hypothetical Syllogism (HS)

$$\begin{array}{l} p \supset q \\ q \supset r \\ \therefore p \supset r \end{array}$$

### 4. Disjunctive Syllogism (DS)

$$\begin{array}{l} p \vee q \\ \sim p \\ \therefore q \end{array}$$

### 5. Constructive Dilemma (CD)

$$\begin{array}{l} (p \supset q) \cdot (r \supset s) \\ p \vee r \\ \therefore q \vee s \end{array}$$

### 6. Simplification (Simp)

$$\begin{array}{l} p \cdot q \\ \therefore p \end{array}$$

### 7. Conjunction (Conj)

$$\begin{array}{l} p \\ q \\ \therefore p \cdot q \end{array}$$

### 8. Addition (Add)

$$\begin{array}{l} p \\ \therefore p \vee q \end{array}$$

## 10 RULES OF REPLACEMENT

### 9. De Morgan's Theorems (DM)

$$\begin{array}{ll} \sim (p \cdot q) & :: \sim p \vee \sim q \\ \sim (p \vee q) & :: \sim p \cdot \sim q \end{array}$$

### 10. Association (Assoc)

$$\begin{array}{ll} p \vee (q \vee r) & :: (p \vee q) \vee r \\ p \cdot (q \cdot r) & :: (p \cdot q) \cdot r \end{array}$$

### 11. Double Negation (DN)

$$p \quad :: \quad \sim \sim p$$

### 12. Material Implication (Impl)

$$p \supset q \quad :: \quad \sim p \vee q$$

### 13. Exportation (Exp)

$$(p \cdot q) \supset r \quad :: \quad p \supset (q \supset r)$$

### 14. Commutation (Com)

$$\begin{array}{ll} p \vee q & :: q \vee p \\ p \cdot q & :: q \cdot p \end{array}$$

### 15. Distribution (Dist)

$$\begin{array}{ll} p \cdot (q \vee r) & :: (p \cdot q) \vee (p \cdot r) \\ p \vee (q \cdot r) & :: (p \vee q) \cdot (p \vee r) \end{array}$$

### 16. Transposition (Trans)

$$p \supset q \quad :: \quad \sim q \supset \sim p$$

### 17. Material Equivalent (Equiv)

$$\begin{array}{ll} p \equiv q & :: (p \supset q) \cdot (q \supset p) \\ p \equiv q & :: (p \cdot q) \vee (\sim p \cdot \sim q) \end{array}$$

### 18. Tautology (Taut)

$$\begin{array}{ll} p & :: p \vee p \\ p & :: p \cdot p \end{array}$$

SAMPLE